

Migraine

Migraine is not merely a severe headache. It is a primary headache disorder characterized by recurrent attacks accompanied by neurological, gastrointestinal, and autonomic changes. Attacks typically last 4–72 hours and can significantly affect daily life. Headache disorders affect approximately 40% of the population worldwide; migraine occurs in approximately 15–20% of women and 6–10% of men. Its higher prevalence in women is attributed to hormonal changes. A significant proportion of patients do not receive appropriate diagnosis and treatment, which may lead to loss of productivity and a reduced quality of life. According to the World Health Organization (WHO), migraine represents a substantial disease burden among neurological disorders due to the functional impairment and disruption of daily activities it causes.

Migraine attacks generally progress through four phases: 1) Prodrome (Premonitory Phase): This begins hours or days before the headache. It occurs in 20–60% of patients. Symptoms may include fatigue, difficulty concentrating, neck stiffness, mood changes, an increased desire to sleep, cravings for sweets, or loss of appetite. The patient may experience a feeling that a migraine is coming. 2) Aura: This occurs in approximately one-third of patients. It lasts 5–60 minutes and usually develops before or during the headache. Visual symptoms are the most common and may include flashes of light, zigzag lines, spots, or blurred vision. Sensory symptoms (tingling, numbness), speech disturbances, or motor symptoms may also occur. These symptoms are completely reversible. 3) Headache: The headache is moderate to severe, often unilateral, and throbbing. It is aggravated by physical activity. Nausea, vomiting, and sensitivity to light and sound commonly accompany the headache. It tends to begin in the morning. 4) Postdrome (Resolution Phase): After the pain subsides, patients may experience fatigue, exhaustion, difficulty concentrating, or sometimes a feeling of being refreshed. This phase continues until the attack has completely resolved.

The most common type is migraine without aura. Migraine with aura is classically recognized as the traditional form of migraine. Other types include chronic migraine (headache occurring on more than 15 days per month, with migraine features on at least 8 of those days), hemiplegic migraine, migraine with brainstem aura, and childhood periodic syndromes (such as cyclic vomiting and abdominal migraine). Medication overuse is an important risk factor for chronification.

Migraine typically begins during adolescence or young adulthood, and most patients experience their first attack before the age of 30. Genetic susceptibility is high, with 60–80% of patients having a family history. Triggers vary from person to person; some of the most common include stress, insufficient or excessive sleep, fasting, certain foods (chocolate,

cheese, citrus fruits), alcohol, changes in caffeine consumption, hormonal fluctuations, weather changes, strong odors, and physical exertion.

Diagnosis is based almost entirely on the patient's history. Laboratory tests or imaging studies (MRI, CT) are performed only when another underlying condition is suspected. Keeping a headache diary is very useful: the time of the attack, its severity, associated symptoms, triggers, and medications taken should be recorded. For migraine without aura, at least 5 attacks and specific characteristics (unilateral location, pulsating quality, moderate-to-severe intensity, and aggravation by physical activity), together with nausea or sensitivity to light and sound, are sought. For migraine with aura, at least 2 attacks with fully reversible neurological symptoms are sufficient.

Treatment is divided into two main approaches: acute treatment and preventive treatment.

1) During an attack: Early intervention is important. For mild attacks, simple analgesics such as paracetamol, ibuprofen, or aspirin may be tried. For moderate-to-severe attacks, triptans are preferred; antiemetics may be added when nausea is present. Resting in a quiet, dark environment, applying a cold compress, and maintaining adequate fluid intake may provide additional relief.

Opioid- and barbiturate-containing medications should be avoided because they increase the risk of chronification.

2) Preventive treatment: Preventive therapy should be considered when a patient experiences 4 or more attacks per month, significant disability, or an inadequate response to acute treatment. Lifestyle modification is fundamental: regular sleep, regular meals, adequate hydration, moderate exercise, stress management, and avoidance of individual triggers. Pharmacological options include beta-blockers (propranolol), antidepressants (amitriptyline), antiepileptic drugs (topiramate), or calcium channel blockers. For chronic migraine, options such as botulinum toxin and CGRP monoclonal antibodies may be considered under neurological supervision.

Behavioral measures, including regular sleep and exercise schedules and relaxation techniques, are recommended for every patient. Treatment should be individualized, taking pregnancy, comorbidities, and patient preferences into account.

In summary, migraine is a potentially lifelong but manageable condition. With accurate diagnosis, early treatment of attacks, identification of triggers, and preventive approaches when necessary, quality of life can be significantly improved. Most cases can be managed in primary care, while patients with complex or refractory disease should be referred to neurology. Patient education and a headache diary are among the most valuable tools for both diagnosis and follow-up.

References:

- 1) Ankara University. *Migren*. Dan. Öğr. Gör. Yar. Doç. Dr. Selda Tekiner, Haz. Araş. Gör. Dr. Ahmet Mekin Çetin. <https://ailehekimimedicine.ankara.edu.tr/wp-content/uploads/sites/581/2015/02/Migren.pdf>
- 2) World Health Organization. (2025, October 24). *Headache disorders*. <https://www.who.int/news-room/fact-sheets/detail/headache-disorders>
- 3) National Institute of Neurological Disorders and Stroke. (2026, August 10). *Migraine*. National Institutes of Health. <https://www.ninds.nih.gov/health-information/disorders/migraine>
- 4) International Headache Society. (2018). *The International Classification of Headache Disorders, 3rd edition (ICHD-3)*. Cephalalgia, 38(1), 1–211. <https://ichd-3.org/1-migraine/>
- 5) Aguilar-Shea AL, Membrilla MdJA, Diaz-de-Teran J. Migraine review for general practice. *Aten Primaria*. 2022;54(2):102208. doi:10.1016/j.aprim.2021.102208. PMID:34798397; PMCID:PMC8605054.

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